

CPT-WORLD: ACTIVE CAUSAL REASONING IN HIDDEN FINITE WORLDS

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ABSTRACT

Causal reasoning begins with a choice of evidence. A capable agent decides which intervention to perform, which variables to read, and when the collected evidence supports a causal conclusion. Current language-model evaluations primarily score answers to supplied causal descriptions, while broad scientific environments mix causal inference with domain scripts and long-horizon control. We introduce CPT-WORLD, an executable benchmark substrate for active causal reasoning in hidden, finite causal worlds. A model receives opaque variables, a causal query, a legal experiment interface, and a structured terminal schema. It gathers selected joint counts through passive observation or batched hard interventions. One exact engine generates the feedback and verifies five terminal objects: total effects, counterfactual transition sets, experimental decisions, minimal adjustment sets, and mediator structure. Counterfactual scoring ranges over every cross-world coupling compatible with the CPT world, so equally compatible structural causal models receive the same legal answer set. Exact variable elimination and ancestral sampling preserve a full-joint rational oracle while accelerating a 15-node sparse benchmark by $338.4\times$ for selected marginals and $102.5\times$ for batch feedback. The resulting environment isolates a central scientific skill: using a controlled experiment budget to recover causal structure and effects from a hidden mechanism.

1 INTRODUCTION

A causal question often arrives before the decisive data. A scientist can observe the system, perturb a controllable variable, select a measurement panel, inspect the outcome, and adapt the next experiment. The quality of the conclusion depends on the whole loop. An agent that computes a correct estimand from a completed data table exercises one part of this capability. An agent that chooses the evidence also exercises causal search, measurement selection, and experimental economy.

Language-model causal benchmarks have made formal causal inference measurable. CLadder constructs identifiable graph, query, and data triples, then verbalizes the causal model and sufficient quantities for a static answer (Jin et al., 2023). Causal experimental-design methods choose interventions that inform a graph or target query (Toth et al., 2022; Tigas et al., 2023), including settings where the treatment can only be perturbed indirectly (Ailer et al., 2023; Dern et al., 2024). Scientific-agent environments add long-horizon interaction across rich domains (Jansen et al., 2024). These lines establish the ingredients for a focused evaluation object: an LLM faces a hidden causal mechanism, controls a finite experimental interface, and receives an exact verdict on the resulting causal answer.

CPT-WORLD realizes this object with finite causal Bayesian networks. The world is a DAG and an exact conditional probability table for every node. The prompt replaces internal variable names with opaque tokens and keeps the graph, CPT entries, truth, and scorer sealed. It exposes legal interventions, readable variables, a maximum readout width, and the terminal JSON schema. The model can request a passive batch or a hard intervention, together with a selected readout. The environment returns sparse joint counts for that readout and preserves every other variable as hidden state. Figure 1 shows the complete loop.

This interaction creates a direct test of experimental causal reasoning. Consider an ATE query whose treatment and outcome are read-only. The model must use legal probes on other variables and decide which endpoints to measure together. A decision query separates exploratory interventions from the final deployment action. Discovery queries ask for minimal adjustment sets or mediator order after

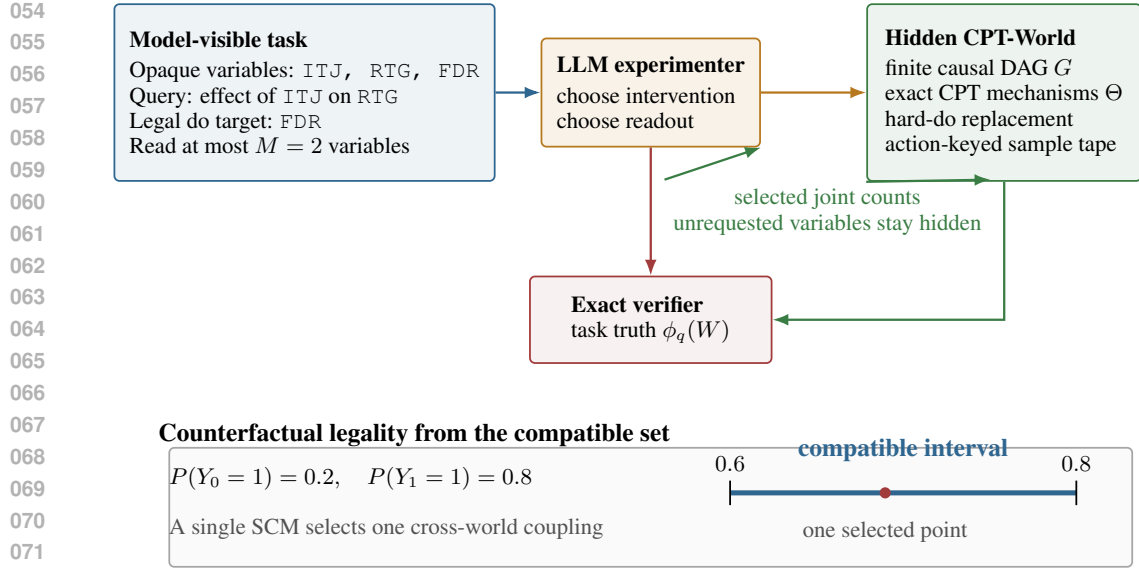


Figure 1: **An experimental causal episode.** The model sees an opaque task and a legal experiment surface. It chooses both a perturbation and a readout, then receives only the requested joint counts. The exact verifier evaluates the terminal causal object. The lower strip shows the counterfactual rule: CPT marginals define a compatible interval, while a single SCM selects one coupling inside that interval.

the model has manipulated and observed the hidden system. The same renderer and runtime support every task, so task variation comes from the causal query and the interaction surface.

Counterfactual evaluation receives a second benefit from the finite CPT representation. A CPT world fixes each interventional marginal. It leaves the cross-world coupling open. If $P(Y_0=1) = 0.2$ and $P(Y_1=1) = 0.8$, then the fraction of units that change from 0 to 1 lies in $[0.6, 0.8]$. A generated SCM selects one point in this interval. That choice injects an evaluator preference into the label and into any training signal derived from it. CPT-WORLD keeps the full compatible set and scores either its sharp interval or a submitted compatible value. The task therefore measures counterfactual reasoning supported by the stated causal information (Balke & Pearl, 1997; Pearl, 2009).

The work contributes a unified experimental causal interface, a five-task construction pipeline, and an exact scalable verifier. The construction samples a world and query first, establishes family-level answerability, then samples the legal intervention subset K and observation width M . The verifier uses exact rational variable elimination for query marginals and action-keyed ancestral sampling for feedback. Differential tests match the original full-joint semantics, and end-to-end tests exercise every task from prompt through terminal score.

2 RELATED WORK AND BENCHMARK POSITION

Causal reasoning in language models. CLadder grounds natural-language questions in symbolic causal graphs, queries, and oracle answers across association, intervention, and counterfactual levels (Jin et al., 2023). Its generator provides the model specification and the quantities needed for an identifiable answer. This design measures formal inference through language and supplies a strong static anchor for our task semantics. CPT-WORLD moves the evidence acquisition step into the episode. The model receives an action interface and creates the data on which its answer rests.

Causal experiment design. ABCI integrates posterior inference over causal models and target queries with sequential intervention selection (Toth et al., 2022). Differentiable multi-target design optimizes batches of target-state pairs (Tigas et al., 2023). A-ICP selects experiments that expose stable parent sets (Gamella & Heinze-Deml, 2020). Sequential instrument selection and targeted indirect design address settings in which the treatment itself is unavailable for manipulation (Ailer

et al., 2023; Dern et al., 2024). CPT-WORLD converts these experimental intuitions into a model-facing benchmark protocol with opaque symbols, discrete tool calls, selected measurements, and task-specific exact scoring.

Scientific interaction environments. DiscoveryWorld evaluates complete cycles of hypothesis formation, experiment design, analysis, and action in simulated scientific domains (Jansen et al., 2024). Its rich environments test broad scientific agency. CPT-WORLD concentrates the environment semantics on causal variables, interventions, and probabilities. This concentration supports exact enumeration of task truth, controlled sampling of the evidence surface, and deterministic replay of stochastic experiments.

Setting	Hidden mechanism	Agent experiments	Selected readout	Exact task oracle
Static causal QA				✓
Causal experiment design	✓	✓	method-specific	✓
Broad science environments	✓	✓	✓	task-specific
CPT-WORLD	✓	✓	✓	✓

Table 1: **Benchmark position.** CPT-WORLD combines a hidden mechanism, model-chosen experiments, bandwidth-limited measurements, and an exact causal oracle in one compact environment.

3 THE CPT-WORLD EPISODE

3.1 A HIDDEN FINITE CAUSAL WORLD

A world is $W = (G, \Theta)$, where G is a finite DAG and Θ contains one exact categorical CPT row for every parent assignment. For variables $V_1, \dots, V_n$ in topological order,

$$P_W(v_1, \dots, v_n) = \prod_{j=1}^n P_{\Theta_j}(v_j \mid v_{\text{pa}(j)}). \quad (1)$$

Hard intervention $\text{do}(V_j = x)$ replaces the mechanism for V_j with a point mass at x . This gives one semantic owner for passive and interventional data (Pearl, 2009; Peters et al., 2017).

The renderer samples opaque three-letter labels and public state tokens. The model receives variable domains, the query, legal hard-do targets, readable variables, legal batch sizes, and the terminal schema. Graph edges, CPT values, internal names, task truth, and scorer state remain sealed.

3.2 ACTION, MEASUREMENT, AND FEEDBACK

An experiment is a compact triple

$$a_t = \left(\underbrace{\text{do}(X = x) \text{ or observe}}_{\text{perturb}}, \underbrace{R_t}_{\text{read}}, \underbrace{b_t}_{\text{repeat}} \right), \quad |R_t| \leq M. \quad (2)$$

The intervention target belongs to a seed-owned legal set of size K . The readout R_t is a nonempty subset of readable variables and excludes the active intervention target. The environment draws b_t IID units from the same hidden world and reports

$$C_{R_t} \sim \text{Multinomial} \left(b_t, \underbrace{P_W(R_t \mid \text{do}(X = x))}_{\text{selected law}} \right). \quad (3)$$

Only nonzero cells appear in the JSON count map. This sparse representation keeps feedback proportional to the batch and selected domain, while preserving an exact meaning for every omitted cell.

3.3 SAMPLING THE INTERACTION SURFACE

The construction order separates causal content from evidence access. It samples a legal world, a query anchor, and an exact task answer. Family-level answerability groups worlds by their complete legal response signatures and checks that each group shares an accepted terminal answer. The sampler then draws K and M . Conditional on the eligible intervention variables, the width K is uniform and the subset is uniform at that width. The readout width M is sampled independently from 1 through n . A fixed world and query can therefore appear with narrow or broad evidence access while retaining the same causal truth.

4 ONE ENGINE, FIVE CAUSAL TASKS

Table 2 summarizes the implemented tasks. Each task returns one structured terminal object. The exact owner computes its truth from the sealed world, and the public protocol uses visible labels throughout.

Task	Question	Terminal object	Experimental separation
Total effect	Effect of X on Y	scalar ATE	X, Y read-only; probe legal non-endpoints
Counterfactual transition	Units changing outcome across two treatments	sharp interval or compatible scalar	query endpoints read-only
Experimental decision	Deployment value optimizing an outcome event	target-value intervention	deployment variable read-only during evidence collection
Backdoor adjustment	Covariates that block every backdoor path	all minimal sets	manipulate and read candidate covariates
Mediator structure	Variables and consecutive edges on directed $X \rightarrow Y$ paths	mediator set and order	perturb path candidates and read descendants

Table 2: **Five tasks share one interaction loop.** The separation column states how each task assigns distinct roles to evidence actions and the terminal causal object.

4.1 EFFECTS AND COUNTERFACTUAL LEGAL SETS

For treatment states x_1, x_0 and outcome event $Y = y$, the total-effect target is

$$\tau = P_W(Y = y \mid \text{do}(X = x_1)) - P_W(Y = y \mid \text{do}(X = x_0)). \quad (4)$$

The generic interaction keeps X and Y read-only, turning the query into an indirect experimental problem. Passive batches and interventions on legal non-endpoints provide the evidence.

For the transition event $q = P(Y_{x_1} = y, Y_{x_0} \neq y)$, the world fixes the two marginals and induces a compatible set of nonparametric SCMs $\mathcal{C}(W)$. The verifier computes

$$[L, U] = \left[\inf_{\mathcal{M} \in \mathcal{C}(W)} q(\mathcal{M}), \sup_{\mathcal{M} \in \mathcal{C}(W)} q(\mathcal{M}) \right] \quad (\text{all compatible couplings}). \quad (5)$$

For the binary event, these are the sharp Fréchet endpoints $L = \max(0, p_1 - p_0)$ and $U = \min(p_1, 1 - p_0)$, where $p_i = P(Y = y \mid \text{do}(X = x_i))$. The interval mode scores both endpoints continuously. The compatible-value mode accepts any point in the interval and reports its distance to the interval.

4.2 DECISION AND STRUCTURAL DISCOVERY

The experimental-decision task names a deployment variable D , an outcome event, and a minimize or maximize objective. Evidence actions operate on separate legal targets. The terminal answer chooses one value of D , and the verifier reports exact regret against every deployment value. This separation evaluates whether the model can use experiments to support a later action.

The two discovery tasks return causal structure relevant to a named query. Backdoor adjustment returns the full family of inclusion-minimal sets. Mediator discovery returns each variable on a directed treatment-outcome path and every consecutive path edge. Set precision, recall, F1, and exact match preserve partial credit while the exact-match fields retain a strict structural verdict.

5 EXACT AND REPRODUCIBLE EXECUTION

5.1 ONE PROBABILITY SEMANTICS, TWO EXECUTION PATHS

Full-joint enumeration remains the reference oracle. Exact selected marginals use factor elimination with the same rational CPT rows and mechanism replacement. For maximum domain size d , induced width w , and selected width m , its cost is $O(nd^{w+1} + d^m)$. The selected law therefore avoids materializing d^n assignments whenever the graph permits a narrow elimination order.

Interactive feedback uses ancestral sampling. Each unit visits the nodes once in topological order and samples from the active CPT row, giving $O(bnd)$ time for a batch of size b . The output map stores at most $\min(b, d^m)$ occupied cells.

5.2 ACTION-KEYED OUTCOME TAPES

Every random draw is addressed by a versioned key containing the canonical intervention, sample index, and node. Visible labels, requested readout, batch partition, and action interleaving stay outside this address. The same causal experiment therefore receives the same latent units when a renderer permutes symbols or a policy divides one batch into several calls. This common-randomness contract supports paired model comparisons and exact replay.

6 VALIDATION

6.1 SEMANTIC TESTS

Table 3 records the validation layers. Exact differential tests compare selected variable elimination with full-joint enumeration over random multi-valued worlds, natural distributions, and hard interventions. Ancestral sampling tests compare every nonzero assignment interval with the full-joint law. Protocol tests cover strict visible labels, selected projection, sparse counts, budget updates, tape invariance, and terminal parsing. The complete suite contains 69 passing tests, and both Ruff checks pass.

Layer	Test	Result
Exact inference	24 random multi-valued worlds under natural and hard-do laws	Fraction-equal to full joint
Batch semantics	12 random DAGs, every nonzero assignment interval	identical categorical mass
Reproducibility	rename, projection, split-batch, and interleaved actions	invariant tape draws
Task loop	prompt, batch feedback, terminal parser, exact scorer	all five modes execute
Repository suite	unit tests, lint, and format	69 pass; checks clean

Table 3: **Validation targets semantic ownership.** Each accelerated or interactive component is compared with the exact reference law or a deterministic protocol invariant.

6.2 END-TO-END TASK DEMONSTRATION

The runtime demonstration samples one instance per task, renders its opaque prompt, executes a legal selected-measure intervention, returns sparse counts, and sends an exact terminal object through the public parser and scorer. A representative ATE episode exposes variables `ITJ`, `RTG`, and `FDR`; keeps the first two read-only; and permits intervention on `FDR`. The command

```
{ "type": "intervene", "target": "FDR", "value": "state.1",  
  "measure": ["RTG", "ITJ"], "batch_size": 8 }
```

returns three occupied joint-count cells totaling eight samples. The terminal ATE is then checked against the exact value $316/479$. The other four modes traverse the same state machine and return their own typed terminal objects.

6.3 SCALING THE EXACT VERIFIER

Figure 2 compares the accelerated paths with the original full-joint implementation on sparse binary chains. Every timed marginal first passes exact Fraction equality. At 10 nodes, variable elimination is $13.8\times$ faster and ancestral feedback is $7.3\times$ faster. At 15 nodes, the speedups reach $338.4\times$ and $102.5\times$. The growth reflects the shift from the joint state count 2^n to graph width for exact queries and to *bnd* work for sampled feedback.

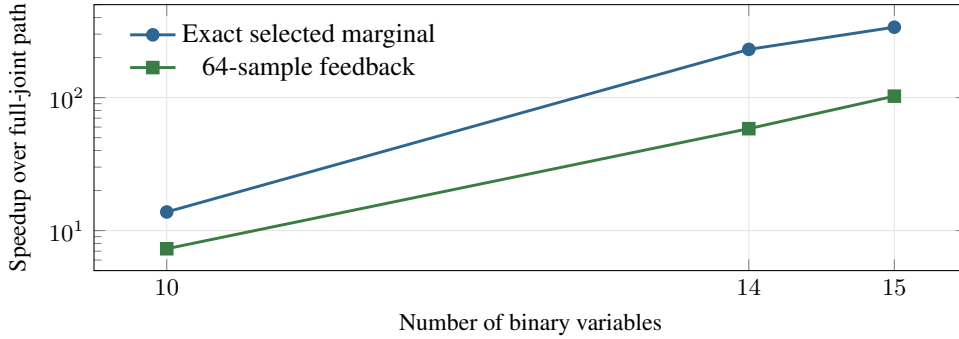


Figure 2: **Exact and interactive acceleration.** Median speedup over the full-joint reference on sparse binary chains. The exact marginal curve uses variable elimination; the feedback curve generates 64 IID units with ancestral sampling.

7 CONCLUSION

CPT-WORLD defines causal reasoning as an experiment-and-answer loop inside a hidden finite world. The model controls interventions and measurements through a compact JSON interface; the environment returns only selected evidence; and one exact semantic layer verifies five causal objects. Compatible-SCM counterfactual sets keep evaluation aligned with the causal information supplied by the world. Exact acceleration and action-keyed replay make the same construction practical for benchmark generation, model evaluation, and training trajectories. This turns finite CPT worlds into an executable laboratory for measuring how models acquire and use causal evidence.

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A FORMAL WORLD SEMANTICS

Let each node V_j have domain $\{0, \dots, d_j - 1\}$. The parent tuple follows a fixed order, so every CPT row has a canonical index. A hard intervention map $I = \{j \mapsto x_j\}$ induces

$$P_W(v \mid \text{do}(I)) = \prod_{j \notin I} P_{\Theta_j}(v_j \mid v_{\text{pa}(j)}) \prod_{j \in I} \mathbf{1}[v_j = x_j]. \quad (6)$$

All probabilities are stored as exact rational numbers. Natural observation is the special case $I = \emptyset$.

The public surface contains a visible permutation σ of node labels, a manipulability mask, a readability mask, and observation width M . Parsing maps visible tokens back to canonical indices before executing any probability operation. Feedback maps selected indices back through σ and serializes only nonzero cells.

B TASK TRUTH AND DIAGNOSTICS

B.1 TOTAL EFFECT

For the rendered treatment values x_1, x_0 and outcome state y , the truth owner evaluates Eq. 4 with exact projected interventional marginals. The terminal diagnostics are absolute and squared error.

B.2 COUNTERFACTUAL TRANSITION

Define $A = \mathbf{1}[Y_{x_1} = y]$ and $B = \mathbf{1}[Y_{x_0} = y]$. The CPT world fixes $P(A = 1) = p_1$ and $P(B = 1) = p_0$. The desired event is $A = 1, B = 0$. Its sharp coupling bounds are

$$\max(0, p_1 - p_0) \leq P(A = 1, B = 0) \leq \min(p_1, 1 - p_0). \quad (7)$$

The interval terminal reports endpoint errors. The compatible-scalar terminal reports membership and distance to the interval. Both modes share the same world, query anchors, visible labels, K , and M .

B.3 EXPERIMENTAL DECISION

For deployment variable D , admissible states $\mathcal{D}$, outcome event $Y = y$, and objective $s \in \{\min, \max\}$, the oracle enumerates $P(Y = y \mid \text{do}(D = d))$ for $d \in \mathcal{D}$. Every optimal state receives zero regret. The scorer computes the exact gap between the chosen probability and the objective optimum.

B.4 BACKDOOR ADJUSTMENT

The structural oracle enumerates observed node sets that satisfy the backdoor criterion for the named treatment and outcome, then retains inclusion-minimal sets. The scorer treats the family of minimal sets as the prediction universe and reports precision, recall, F1, and exact family match.

B.5 MEDIATOR STRUCTURE

The oracle enumerates directed treatment-outcome paths. The mediator set is the union of internal path nodes. The order relation contains each consecutive edge on at least one such path. Mediator and edge sets receive separate precision, recall, F1, and exact-match diagnostics.

C EXACT VARIABLE ELIMINATION

For a requested marginal R , each CPT becomes a factor over one node and its parents. A hard intervention replaces the corresponding factor with a point mass. The engine eliminates non-readout variables in a min-fill order and multiplies the remaining factors before canonical normalization. With induced width w , maximum domain size d , and readout width m , the upper cost is $O(nd^{w+1} + d^m)$ time. The reference implementation enumerates every assignment and remains available for differential testing.

D ANCESTRAL SAMPLING AND REPLAY

For each sample index i and canonical node j , the tape produces a rational uniform draw addressed by

$$\text{key} = (\text{version}, \text{tape id}, I, i, j). \quad (8)$$

The sampler visits nodes in topological order, uses intervention values for nodes in I , and inverse-samples the active CPT row for every other node. It projects the full latent assignment to the requested readout only after the sample is complete. A batch split preserves the same sample indices. An interleaved action uses an independent intervention arm address.

The old full-joint batch path costs $O((n+b)d^n)$ time and $O(d^n + d^m)$ memory. The ancestral path costs $O(bnd)$ time and $O(n + \min(b, d^m))$ working and output memory.

E CONSTRUCTION PIPELINE

The implemented generator follows this order:

1. sample a node count from the configured support;
2. sample each finite domain size, a topological order, and a forward-edge subset;
3. sample exact rational CPT parameters;
4. enumerate legal query anchors and construct exact task truth;
5. evaluate answerability over the complete query-specific experiment surface;
6. sample the legal intervention subset K and readout width M ;
7. sample opaque labels and render the visible episode.

The reference configuration uses node counts $\{2, 3, 4\}$ and domains of two through five states. The probability engine is exercised separately through 15 binary nodes in the scaling benchmark.

Family-level answerability uses an exact response signature. Two candidate worlds share a signature when every legal passive and hard-do projected distribution is equal on the complete family surface. Effect and structural tasks require equal answers inside each signature class. Decision tasks require a shared zero-regret deployment action. K and M are sampled after this partition.

F MODEL-VISIBLE PROTOCOL

The initial message follows a fixed shape:

```
CPT-WORLD HIDDEN-MECHANISM TASK
Variables and finite states: [opaque labels and state
tokens]
Query: [typed causal question]
Legal experimental do targets: [visible labels]
measure must be a nonempty subset of readable variables
At most M variables may be measured per batch
To finish, return exactly one typed JSON object
```

An intervention command is

```
{ "type": "intervene", "target": "LAB",
  "value": "state_i", "measure": ["LAB", ...], "batch_size": n }.
```

A passive command replaces the target and value fields with `"type": "observe"`. The parser rejects hidden internal labels, noncanonical state tokens, duplicate JSON keys, illegal targets, duplicate or unreadable measurements, target-in-measure requests, and unsupported batch sizes.

G VALIDATION AND BENCHMARK SETUP

The exact marginal differential covers 24 random multi-valued worlds, both the natural law and hard interventions, and compares every projected cell as a `Fraction`. The ancestral test covers 12 random DAGs and checks the cumulative interval assigned to every nonzero full assignment. Runtime tests cover selected joint projection, sparse serialization, passive observation, hard-do replacement, budget state, terminal dispatch, and all five query modes.

The acceleration benchmark uses a binary chain. The root is Bernoulli $1/2$ and each transition preserves the parent state with probability $3/4$. Both paths compute the last-node marginal under $\text{do}(V_0 = 1)$. The batch comparison uses 64 samples and the same action-keyed tape. Each exact timing run first checks equality with the full-joint reference.